

ADVERTISEMENT

lansoprazole (Rx, OTC)

Brand and Other Names: Prevacid, Prevacid Solu Tab, Prevacid 24HR
Classes: Proton Pump Inhibitors

▼

Dosing & Uses

ADULT

Dosage Forms & Strengths

capsule delayed release

- 15mg
- 30mg

tablet, oral-disintegrating

- 15mg
- 30mg

Duodenal Ulcer

Active: 15 mg PO qDay for 4 weeks

Maintenance: 15 mg PO qDay

Gastric Ulcer

30 mg PO qDay for 8 weeks

ADVERTISEMENT

NSAID-associated GU

Treatment: 30 mg PO qDay for 8 weeks

Prevention: 15 mg PO qDay for 12 weeks

Gastroesophageal Reflux Disease

15 mg PO qDay for 8 weeks

Erosive Esophagitis

30 mg PO qDay for 8-16 weeks

Maintenance: 15 mg PO qDay

Hypersecretory Condition (eg, Zollinger-Ellison Syndrome)

60 mg PO qDay initially; up to 180 mg q12hr used

If dose >120 mg/day PO, administer in divided doses q12hr

Helicobacter Pylori Infection

Triple therapy: Lansoprazole 30 mg + amoxicillin 1 g + clarithromycin 500 mg PO q12hr for 10-14 days

Dual therapy (clarithromycin resistant): Lansoprazole 30 mg + amoxicillin 1 g PO q8hr for 14 days

Penicillin allergy: Lansoprazole 30 mg + clarithromycin 500 mg + metronidazole 500 mg q12hr for 10-14 days

Heartburn

OTC product: 15 mg PO qDay for 14 days; may repeat q4Months

Dosage Modifications**Hepatic impairment**

- Mild (Child-Pugh Class A) or moderate (Child-Pugh Class B) hepatic impairment: No dosage adjustment necessary
- Severe hepatic impairment (Child-Pugh Class C): 15 mg PO qDay

PEDIATRIC**Dosage Forms & Strengths**

capsule delayed release

- 15mg
- 30mg

tablet, oral-disintegrating

- 15mg
- 30mg

GERD, Erosive Esophagitis**<1 year**

- Safety and efficacy not established

1-12 years

- <30 kg: 15 mg PO qDay for 8-12 weeks
- >30 kg: 30 mg PO qDay for 8-12 weeks
- May increase dose to 30 mg PO q12hr after >2 weeks of initial therapy if still symptomatic

>12 years

- 30 mg PO qDay for up to 8 weeks

Nonerosive GERD

<12 years: Safety and efficacy not established

≥12 years: 15 mg PO qDay for up to 8 weeks

Interactions

Interaction Checker

Enter a drug name to check for any interactions. and

lansoprazole

No Interactions Found

Interactions Found

Contraindicated

Serious

Significant - Monitor Closely

Minor

All Interactions Sort By: Severity ▼

Contraindicated (2)

- **nelfinavir**
lansoprazole decreases levels of nelfinavir by unspecified interaction mechanism. Contraindicated. Coadministration may lead to loss of nelfinavir virologic response and development of resistance; mechanism may be CYP2C19 inhibition of nelfinavir conversion to active M8 metabolite, and also PPIs decreasing gastric pH resulting in decreased nelfinavir absorption.
- **rilpivirine**
lansoprazole decreases levels of rilpivirine by increasing gastric pH. Applies only to oral form of both agents. Contraindicated. Concurrent use may cause treatment failure and/or the development of rilpivirine or NNRTI resistance owing to decreased levels.

Serious (34)

- **acalabrutinib**
lansoprazole decreases levels of acalabrutinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Acalabrutinib solubility decreases with increasing gastric pH. Due to the long-lasting effect of PPIs, separation of doses may not eliminate the interaction.
- **apalutamide**
apalutamide will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Coadministration of apalutamide, a strong CYP2C19 inducer, with drugs that are CYP2C19 substrates can result in lower exposure to these medications. Avoid or substitute another drug for these medications when possible. Evaluate for loss of therapeutic effect if medication must be coadministered.

apalutamide will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Coadministration of apalutamide, a strong CYP3A4 inducer, with drugs that are CYP3A4 substrates can result in lower exposure to these medications. Avoid or substitute another drug for these medications when possible. Evaluate for loss of therapeutic effect if medication must be coadministered. Adjust dose according to prescribing information if needed.
- **atazanavir**
lansoprazole will decrease the level or effect of atazanavir by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Atazanavir solubility decreases as pH increases. Substantially reduced plasma concentrations of atazanavir are expected if PPIs are coadministered. PPI dose should not exceed a dose comparable to omeprazole 20 mg and must be taken ~12 h before atazanavir/ritonavir in treatment naive-patients. PPIs are not recommended in treatment-experienced taking atazanavir.
- **dacomitinib**

lansoprazole will increase the level or effect of dacomitinib by unspecified interaction mechanism. Avoid or Use Alternate Drug. Concomitant use with a PPI decreases dacomitinib concentrations, which may reduce dacomitinib efficacy. Avoid use of PPIs with dacomitinib. As an alternative to PPIs, use locally-acting antacids or an H2-receptor antagonist. Administer at least 6 hours before or 10 hours after taking an H2-receptor antagonist.

- dapsone
lansoprazole will decrease the level or effect of dapsone by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- dasatinib
lansoprazole will decrease the level or effect of dasatinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- defactinib
lansoprazole decreases levels of defactinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Defactinib exposure may be decreased when coadministered with proton pump inhibitors (PPIs). If concomitant use with PPIs cannot be avoided, separate administration by at least 2 hr.
- digoxin
lansoprazole will increase the level or effect of digoxin by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- enzalutamide
enzalutamide will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug.
- erlotinib
lansoprazole decreases levels of erlotinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration of proton pump inhibitors and erlotinib may decrease erlotinib exposure and reduce its efficacy. Consider alternate acid reducing agents.
- fedratinib
lansoprazole will increase the level or effect of fedratinib by Other (see comment). Avoid or Use Alternate Drug. Avoid coadministration of fedratinib (a CYP3A4 and CYP2C19 substrate) with dual CYP3A4 and CYP2C19 inhibitor. Effect of coadministration of a dual CYP3A4 and CYP2C19 inhibitor with fedratinib has not been studied.
- idelalisib
idelalisib will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Idelalisib is a strong CYP3A inhibitor; avoid coadministration with sensitive CYP3A substrates
- infigratinib (DSC)
lansoprazole will decrease the level or effect of infigratinib (DSC) by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- itraconazole

lansoprazole will decrease the level or effect of itraconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- ivosidenib
ivosidenib will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Avoid coadministration of sensitive CYP3A4 substrates with ivosidenib or replace with alternative therapies. If coadministration is unavoidable, monitor patients for loss of therapeutic effect of these drugs.
- ketoconazole
lansoprazole will decrease the level or effect of ketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- levoketoconazole
lansoprazole will decrease the level or effect of levoketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- lonafarnib
lansoprazole will increase the level or effect of lonafarnib by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. If coadministration of lonafarnib (a sensitive CYP3A substrate) with weak CYP3A inhibitors is unavoidable, reduce to, or continue lonafarnib at starting dose. Closely monitor for arrhythmias and events (eg, syncope, heart palpitations) since lonafarnib effect on QT interval is unknown.

lonafarnib will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Lonafarnib may increase the AUC and peak concentration of CYP2C19 substrates. If coadministration unavoidable, monitor for adverse reactions and reduce the CYP2C19 substrate dose in accordance with its approved product labeling.

- mesalamine
lansoprazole decreases effects of mesalamine by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Applies only to sustained release dosage form.
- neratinib
lansoprazole will decrease the level or effect of neratinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- nilotinib
lansoprazole decreases levels of nilotinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. As an alternative to PPIs, use H2-blockers ~10 hr before or ~2 hr after nilotinib, or use antacids ~2 hr before or ~2 hr after nilotinib (does not apply if using ODT nilotinib).
- nisoldipine
lansoprazole will increase the level or effect of nisoldipine by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- pazopanib

lansoprazole will decrease the level or effect of pazopanib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Avoid coadministration of pazopanib with drugs that raise gastric pH; consider short-acting antacids in place of PPIs and H2 antagonists; separate antacid and pazopanib dosing by several hours

- pexidartinib

lansoprazole will decrease the level or effect of pexidartinib by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Avoid coadministration of proton pump inhibitors (PPIs) with pexidartinib. Use H2-receptor antagonists or antacids if needed. When using alternatives to PPIs, administer pexidartinib 2 hr before or after taking locally-acting antacids OR administer pexidartinib at least 2 hr before or 10 hr after taking an H2-receptor antagonist.

- phenytoin

phenytoin will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Consider alternative for one of the interacting drugs

phenytoin will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Consider alternative for one of the interacting drugs

- rifampin

rifampin will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug.

- rilzabrutinib

lansoprazole decreases levels of rilzabrutinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration may decrease rilzabrutinib exposure, resulting in reduced efficacy.

- secretin

lansoprazole, secretin. Other (see comment). Avoid or Use Alternate Drug. Comment: Concomitant use of PPIs may cause a hyperresponse in gastrin secretion in response to stimulation testing with secretin, falsely suggesting gastrinoma. The time it takes for serum gastrin concentrations to return to baseline following discontinuation of PPIs is specific to the individual PPI. Temporarily stop lansoprazole treatment at least 30 days before assessing to allow gastrin levels to return to baseline.

- sofosbuvir/velpatasvir

lansoprazole will decrease the level or effect of sofosbuvir/velpatasvir by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Velpatasvir solubility decreases as gastric pH increases (practically insoluble at pH >5). Coadministration of sofosbuvir/velpatasvir with omeprazole or other PPIs is not recommended. If considered medically necessary, give sofosbuvir/velpatasvir with food 4 hr before omeprazole 20 mg. Use with other PPIs has not been studied.

- sotorasib

lansoprazole will decrease the level or effect of sotorasib by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. If

use with an acid-reducing agent cannot be avoided, administer sotorasib 4 hr before or 10 hr after administration of a locally-acting antacid.

- sparsentan
lansoprazole decreases effects of sparsentan by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Proton pump inhibitors may decrease sparsentan exposure which may reduce efficacy of sparsentan.
- taletrectinib
lansoprazole decreases levels of taletrectinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration may decrease taletrectinib exposure and reduce its efficacy.
- tucatinib
tucatinib will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Avoid concomitant use of tucatinib with CYP3A substrates, where minimal concentration changes may lead to serious or life-threatening toxicities. If unavoidable, reduce CYP3A substrate dose according to product labeling.
- ziftomenib
lansoprazole decreases levels of ziftomenib by unspecified interaction mechanism. Avoid or Use Alternate Drug. Proton pump inhibitors decrease zifomenib peak plasma concentrations by 70%.

Monitor Closely (78)

- amphetamine
lansoprazole, amphetamine. Other (see comment). Use Caution/Monitor. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .
- ampicillin
lansoprazole will decrease the level or effect of ampicillin by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- avapritinib
lansoprazole will increase the level or effect of avapritinib by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- axitinib
lansoprazole increases levels of axitinib by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- belumosudil
lansoprazole will decrease the level or effect of belumosudil by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Increase belumosudil dosage to 200 mg PO BID when coadministered with proton pump inhibitors.
- bosutinib

lansoprazole decreases levels of bosutinib by Other (see comment). Use Caution/Monitor. Comment: PPIs may decrease bosutinib concentration by ~45%; bosutinib displays pH-dependent solubility.

- budesonide

lansoprazole decreases effects of budesonide by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Enteric-coated budesonide dissolves at pH >5.5. Also, dissolution of extended-release budesonide tablets is pH dependent. Coadministration with drugs that increase gastric pH may cause these budesonide products to prematurely dissolve, and possibly affect release properties and absorption of the drug in the duodenum.

- cannabidiol

lansoprazole will increase the level or effect of cannabidiol by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the cannabidiol dose when coadministered with a moderate CYP2C19 inhibitor.

cannabidiol will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the dose of sensitive CYP2C19 substrates, as clinically appropriate, when coadministered with cannabidiol.

- capecitabine

lansoprazole decreases effects of capecitabine by unknown mechanism. Use Caution/Monitor. Retrospective data suggest elevated gastric pH caused by PPI use may impair capecitabine tablet dissolution and/or reduce absorption.

- carbamazepine

carbamazepine will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

carbamazepine will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- carbonyl iron

lansoprazole will decrease the level or effect of carbonyl iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- cenobamate

cenobamate will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Increase dose of CYP3A4 substrate, as needed, when coadministered with cenobamate.

cenobamate will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider a dose reduction of CYP2C19 substrates, as clinically appropriate, when used concomitantly with cenobamate.

- clarithromycin

clarithromycin will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- clobazam
lansoprazole will increase the level or effect of clobazam by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Dosage adjustment may be required; CYP2C19 inhibitors may result in increased exposure to N-desmethyloclobazam (active metabolite).
- clopidogrel
lansoprazole decreases effects of clopidogrel by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Mean AUC of clopidogrel active metabolite was reduced by ~14% when lansoprazole was coadministered compared to administration of clopidogrel alone in healthy subjects who were CYP2C19 extensive metabolizers. Clopidogrel efficacy may be reduced by drugs that inhibit CYP2C19. Inhibition of platelet aggregation by clopidogrel is entirely due to the active clopidogrel metabolite. Clopidogrel is metabolized in part by CYP2C19.
- cobicistat
cobicistat will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- conivaptan
conivaptan will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- crizotinib
lansoprazole decreases levels of crizotinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that elevate the gastric pH may decrease the solubility of crizotinib and subsequently reduce its bioavailability. However, no formal studies have been conducted. .
- dabrafenib
lansoprazole will decrease the level or effect of dabrafenib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that alter upper GI tract pH (eg, PPIs, H2-blockers, antacids) may decrease dabrafenib solubility and reduce its bioavailability

dabrafenib will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely.
- dextroamphetamine
lansoprazole, dextroamphetamine. Other (see comment). Use Caution/Monitor. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .
- diazepam buccal
lansoprazole will increase the level or effect of diazepam buccal by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Strong or moderate CYP2C19 inhibitors may decrease rate of diazepam elimination, thereby increasing adverse reactions to diazepam.
- diazepam intranasal
lansoprazole will increase the level or effect of diazepam intranasal by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Strong or moderate

CYP2C19 inhibitors may decrease rate of diazepam elimination, thereby increasing adverse reactions to diazepam.

- digoxin
lansoprazole increases toxicity of digoxin by Other (see comment). Use Caution/Monitor. Comment: Prolonged use of PPIs may cause hypomagnesemia and increase risk for digoxin toxicity.
- dordaviprone
dordaviprone and lansoprazole both increase QTc interval. Use Caution/Monitor. Increase the frequency of ECG and electrolyte monitoring.
- efavirenz
efavirenz will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- elagolix
elagolix will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Elagolix is a weak CYP2C19 inhibitor. Caution with sensitive CYP2C19 substrates.

elagolix decreases levels of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Elagolix is a weak-to-moderate CYP3A4 inducer. Monitor CYP3A substrates if coadministered. Consider increasing CYP3A substrate dose if needed.

- encorafenib
encorafenib, lansoprazole. affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Encorafenib both inhibits and induces CYP3A4 at clinically relevant plasma concentrations. Coadministration of encorafenib with sensitive CYP3A4 substrates may result in increased toxicity or decreased efficacy of these agents.
- eslicarbazepine acetate
eslicarbazepine acetate will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- etravirine
etravirine will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- fedratinib
fedratinib will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Adjust dose of drugs that are CYP3A4 substrates as necessary.

fedratinib will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Adjust dose of drugs that are CYP2C19 substrates as necessary.

- ferric gluconate
lansoprazole will decrease the level or effect of ferric gluconate by increasing

gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferric maltol
lansoprazole will decrease the level or effect of ferric maltol by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ferrous fumarate
lansoprazole will decrease the level or effect of ferrous fumarate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ferrous gluconate
lansoprazole will decrease the level or effect of ferrous gluconate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ferrous sulfate
lansoprazole will decrease the level or effect of ferrous sulfate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- fexinidazole
fexinidazole will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- finerenone
lansoprazole will increase the level or effect of finerenone by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Monitor serum potassium during initiation and dosage adjustment of either finerenone or weak CYP3A4 inhibitors. Adjust finerenone dosage as needed.
- flibanserin
lansoprazole will increase the level or effect of flibanserin by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Increased flibanserin adverse effects may occur if coadministered with multiple weak CYP3A4 inhibitors.
- fluvoxamine
fluvoxamine will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- fosamprenavir
lansoprazole will decrease the level or effect of fosamprenavir by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- gefitinib
lansoprazole decreases levels of gefitinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Avoid coadministration of gefitinib with PPIs if possible. If treatment with a PPI is required, separate gefitinib and PPI doses by 12 hr.
- iron dextran complex
lansoprazole will decrease the level or effect of iron dextran complex by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- iron sucrose

lansoprazole will decrease the level or effect of iron sucrose by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- isavuconazonium sulfate
lansoprazole will increase the level or effect of isavuconazonium sulfate by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- isoniazid
isoniazid will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- istradefylline
istradefylline will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Istradefylline 40 mg/day increased peak levels and AUC of CYP3A4 substrates in clinical trials. This effect was not observed with istradefylline 20 mg/day. Consider dose reduction of sensitive CYP3A4 substrates.
- itraconazole
itraconazole will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Separate proton pump inhibitors at least 2 hr before or 2 hr after itraconazole. Use of Sporanox oral solution (instead of the capsule) or administration of itraconazole with an acidic beverage (eg, cola) may minimize the significance of this interaction. Closely monitor for reduced itraconazole efficacy if combined.
- ketoconazole
ketoconazole will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- ledipasvir/sofosbuvir
lansoprazole decreases levels of ledipasvir/sofosbuvir by Other (see comment). Use Caution/Monitor. Comment: Ledipasvir solubility decreases as pH increases; drugs that increase gastric pH are expected to decrease levels of ledipasvir; proton-pump inhibitor doses comparable to omeprazole <20 mg can be administered simultaneously with ledipasvir/sofosbuvir under fasted conditions.
- lemborexant
lansoprazole will increase the level or effect of lemborexant by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Lower nightly dose of lemborexant recommended if coadministered with weak CYP3A4 inhibitors. See drug monograph for specific dosage modification.
- lenacapavir
lenacapavir will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Lenacapavir may increase CYP3A4 substrates initiated within 9 months after last SC dose of lenacapavir, which may increase potential risk of adverse reactions of CYP3A4 substrates.
- letermovir
letermovir increases levels of lansoprazole by affecting hepatic enzyme CYP2E1

metabolism. Use Caution/Monitor.

- levoketoconazole
levoketoconazole will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- lorlatinib
lorlatinib will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- lumacaftor/ivacaftor
lumacaftor/ivacaftor, lansoprazole. affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. In vitro studies suggest that lumacaftor may induce and ivacaftor may inhibit CYP2C19 substrates. .

lumacaftor/ivacaftor will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
Lumacaftor/ivacaftor is a strong CYP3A4 inhibitor and also has the potential to induce CYP2C19 and both induce and inhibitor P-gp.

- methotrexate
lansoprazole increases levels of methotrexate by decreasing renal clearance. Use Caution/Monitor. Increased risk of toxicity with higher doses.
- methylphenidate
lansoprazole decreases effects of methylphenidate by enhancing GI absorption. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Since the characteristics of methylphenidate extended release capsules (Ritalin LA) are pH dependent, coadministration of antacids or acid suppressants could alter the release of methylphenidate. Consider separating the administration of the antacid and the methylphenidate extended-release capsules may be avoided.
- midazolam intranasal
lansoprazole will increase the level or effect of midazolam intranasal by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
Coadministration of mild CYP3A4 inhibitors with midazolam intranasal may cause higher midazolam systemic exposure, which may prolong sedation.
- mitapivat
mitapivat decreases levels of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Monitor patients for loss of therapeutic effect if taking sensitive CYP2C19 substrates. .
- mitotane
mitotane decreases levels of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Mitotane is a strong inducer of cytochrome P-4503A4; monitor when coadministered with CYP3A4 substrates for possible dosage adjustments.
- mycophenolate
lansoprazole will decrease the level or effect of mycophenolate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Potential

interaction applies to mycophenolate mofetil. Enteric coated mycophenolate sodium formulation is less sensitive to this interaction. Clinical significance unclear.

- nefazodone
nefazodone will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- paltusotine
lansoprazole decreases levels of paltusotine by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Coadministration demonstrated a dose-dependent decrease in paltusotine exposure owing to decreased paltusotine solubility at elevated gastric pH. May require increased paltusotine dose. Avoid coadministration if patient is already taking paltusotine 60 mg.
- phenobarbital
phenobarbital will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- polysaccharide iron
lansoprazole will decrease the level or effect of polysaccharide iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- posaconazole
lansoprazole will decrease the level or effect of posaconazole by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

posaconazole will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- primidone
primidone will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- revumenib
revumenib and lansoprazole both increase QTc interval. Use Caution/Monitor. ECG monitoring recommended if concomitant use can't be avoided and withhold therapy if QTc interval is >480 msec; restart therapy after QTc interval returns to <480 msec
- rose hips
lansoprazole will decrease the level or effect of rose hips by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- sparsentan
sparsentan will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Sparsentan (a CYP2C19 inducer) decreases exposure of CYP2C19 substrates and reduces efficacy related to these substrates.
- St John's Wort
St John's Wort will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

St John's Wort will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- **stiripentol**
stiripentol, lansoprazole. affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Stiripentol is a CYP3A4 inhibitor and inducer. Monitor CYP3A4 substrates coadministered with stiripentol for increased or decreased effects. CYP3A4 substrates may require dosage adjustment.

stiripentol will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the dose of CYP2C19 substrates, if adverse reactions are experienced when administered concomitantly with stiripentol.

- **tacrolimus**
lansoprazole will increase the level or effect of tacrolimus by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Concomitant administration may increase tacrolimus whole blood concentrations, particularly in intermediate or poor metabolizers of CYP2C19
- **taletrectinib**
taletrectinib and lansoprazole both increase QTc interval. Use Caution/Monitor. Concomitant use may increase the risk of QTc interval prolongation. If concomitant use is unavoidable, adjust the frequency of ECG and electrolyte monitoring.
- **tecovirimat**
tecovirimat will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Tecovirimat is a weak inhibitor of CYP2C8 and CYP2C19. Monitor for adverse effects if coadministered with sensitive substrates of these enzymes.

tecovirimat will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Tecovirimat is a weak CYP3A4 inducer. Monitor sensitive CYP3A4 substrates for effectiveness if coadministered.

- **tinidazole**
lansoprazole will increase the level or effect of tinidazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- **triclabendazole**
triclabendazole will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. If plasma concentrations of the CYP2C19 substrates are elevated during triclabendazole, recheck plasma concentration of the CYP2C19 substrates after discontinuation of triclabendazole.
- **ziftomenib**
lansoprazole, ziftomenib. QTc interval. Use Caution/Monitor. Caution if ziftomenib is coadministered with drugs that exacerbated risk of QT prolongation (eg, hypokalemia, orthostatic hypotension, inhibits metabolism of ziftomenib).

Minor (54)

- acetazolamide
acetazolamide will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- anastrozole
anastrozole will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- armodafinil
armodafinil will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- atazanavir
atazanavir will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- blessed thistle
blessed thistle decreases effects of lansoprazole by pharmacodynamic antagonism. Minor/Significance Unknown. Theoretical interaction.
- bortezomib
bortezomib will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- bosentan
bosentan will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- butalbital
butalbital will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- cyanocobalamin
lansoprazole decreases levels of cyanocobalamin by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.
- cyclophosphamide
cyclophosphamide will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- danazol
danazol will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- darifenacin
darifenacin decreases effects of lansoprazole by Other (see comment). Minor/Significance Unknown. Comment: Effectiveness of proton pump inhibitors may be decreased theoretically if administered with other antisecretory agents.
- dasatinib

dasatinib increases levels of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- devil's claw
devil's claw decreases effects of lansoprazole by pharmacodynamic antagonism. Minor/Significance Unknown.
- dexamethasone
dexamethasone will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- dronedarone
dronedarone will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- drospirenone
drospirenone will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- efavirenz
efavirenz will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- elvitegravir/cobicistat/emtricitabine/tenofovir DF
elvitegravir/cobicistat/emtricitabine/tenofovir DF, lansoprazole. affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown. Based on drug interaction studies conducted with the components of Stribild, no clinically significant drug interactions have been either observed or are expected when coadministered with PPIs.
- erythromycin base
erythromycin base will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- erythromycin ethylsuccinate
erythromycin ethylsuccinate will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- erythromycin lactobionate
erythromycin lactobionate will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- erythromycin stearate
erythromycin stearate will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- etravirine
etravirine will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- ferric carboxymaltose

lansoprazole will decrease the level or effect of ferric carboxymaltose by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown.

- fluconazole

fluconazole will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

fluconazole will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- fluoxetine

fluoxetine will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- fosphenytoin

fosphenytoin will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- larotrectinib

larotrectinib will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- levothyroxine

lansoprazole decreases levels of levothyroxine by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- liothyronine

lansoprazole decreases levels of liothyronine by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- liotrix

lansoprazole decreases levels of liotrix by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- lisdexamfetamine

lansoprazole, lisdexamfetamine. Other (see comment). Minor/Significance Unknown. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .

- methamphetamine

lansoprazole decreases levels of methamphetamine by Other (see comment). Minor/Significance Unknown. Comment: Time to maximum concentration (Tmax) of amphetamine is decreased compared to when administered alone; monitor patients for changes in clinical effect and adjust therapy based on clinical response.

- methylprednisolone

methylprednisolone will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- **modafinil**
modafinil will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown. Net effect on pantoprazole actions unknown due to opposing effects of CYP450 enzymes; monitor
- **nafcillin**
nafcillin will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- **nelfinavir**
nelfinavir will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- **nevirapine**
nevirapine will decrease the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- **oxcarbazepine**
oxcarbazepine will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- **pentobarbital**
pentobarbital will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- **phenobarbital**
phenobarbital will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- **phytoestrogens**
lansoprazole decreases levels of phytoestrogens by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.
- **ponatinib**
lansoprazole decreases levels of ponatinib by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown.
- **primidone**
primidone will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- **ribociclib**
ribociclib will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- **rifampin**
rifampin will decrease the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- **rifapentine**
rifapentine will decrease the level or effect of lansoprazole by affecting

hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- ritonavir
ritonavir will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- ruxolitinib topical
lansoprazole will increase the level or effect of ruxolitinib topical by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- sucralfate
sucralfate decreases levels of lansoprazole by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown. Separate by 30 minutes.
- theophylline
lansoprazole increases levels of theophylline by decreasing metabolism. Minor/Significance Unknown.

lansoprazole increases toxicity of theophylline by Other (see comment). Minor/Significance Unknown. Comment: Prolonged use of proton pump inhibitors can cause hypochlorhydria, which in turn causes peristalsis in small intestine to increase and peristalsis in the proximal colon to decrease; monitor for toxicity.

- thyroid desiccated
lansoprazole decreases levels of thyroid desiccated by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.
- voriconazole
voriconazole will increase the level or effect of lansoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

voriconazole will increase the level or effect of lansoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

Adverse Effects

1-10%

Headache (3-7%)

Diarrhea (1-5%)

Constipation (1-5%)

Nausea (1-3%)

Abdominal pain (1-3%)

<1%

Anxiety

Angina

Palpitations

Syncope

Edema

Anorexia

Dry mouth

Tenesmus

Flatulence

Melena

Myalgia

Tinnitus

Allergic reaction

Postmarketing Reports

Bone fracture

Fundic gland polyps

Acute tubulointerstitial nephritis

Drug reaction with eosinophilia and systemic symptoms (DRESS), acute generalized exanthematous pustulosis (AGEP)

Hypomagnesemia, hypocalcemia, hypokalemia, hyponatremia

Warnings

Contraindications

Hypersensitivity to lansoprazole or other proton pump inhibitors

Coadministration with rilpivirine-containing products

Cautions

Proton pump inhibitors (PPIs) are possibly associated with increased incidence of Clostridium difficile-associated diarrhea (CDAD); consider diagnosis of CDAD for patients taking PPIs who have diarrhea that does not improve

Liver disease may require dosage reduction

May cause severe skin reactions; symptoms may include skin reddening, blisters, rash; if

allergic reaction occurs, stop use and seek medical help right away

Cutaneous lupus erythematosus (CLE) and systemic lupus erythematosus (SLE) reported with PPIs; avoid using for longer than medically indicated; discontinue if signs or symptoms consistent with CLE or SLE are observed and refer patient to specialist

Severe cutaneous adverse reactions, including Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS), and acute generalized exanthematous pustulosis (AGEP) reported in association with use of PPIs; discontinue therapy at first signs or symptoms of severe cutaneous adverse reactions or other signs of hypersensitivity and consider further evaluation

Published observational studies suggest that PPI therapy may be associated with an increased risk for osteoporosis-related fractures of the hip, wrist, or spine, particularly with prolonged (>1 yr), high-dose therapy

Decreased gastric acidity increases serum chromogranin A (CgA) levels and may cause false-positive diagnostic results for neuroendocrine tumors; temporarily discontinue PPIs before assessing CgA levels

PPIs may decrease the efficacy of clopidogrel by reducing the formation of the active metabolite

Relief of symptoms does not eliminate the possibility of a gastric malignancy

Therapy increases risk of Salmonella, Campylobacter, and other infections

Acute interstitial nephritis reported in patients taking proton pump inhibitors; discontinue therapy if interstitial nephritis develops

Daily long-term use (e.g., longer than 3 years) may lead to malabsorption or a deficiency of cyanocobalamin

May elevate and/or prolong serum concentrations of methotrexate and/or its metabolite when administered concomitantly with PPIs, possibly leading to toxicity; consider a temporary withdrawal of PPI therapy with high dose methotrexate administration

SoluTab dosage form contains phenylalanine, a component of aspartame, which can be harmful to patients with phenylketonuria (PKU); before prescribing lansoprazole to a patient with PKU, consider the combined daily amount of phenylalanine from all sources, including the SoluTab dosage form

Some dosage forms contain benzyl alcohol, which has been associated with gasping syndrome in the neonate; avoid or use dosage forms containing benzyl alcohol with caution in neonates

PPI therapy is associated with increased risk of fundic gland polyp; risk increases with long-term use >1 year; patient may be asymptomatic; problem usually identified incidentally on endoscopy; use shortest duration of therapy appropriate to condition being treated

If patient taking a prescription drug, the patient should ask a doctor or a pharmacist whether acid reducers can be taken concomitantly with it

Acute tubulointerstitial nephritis (TIN) reported in patients taking PPIs; may occur at any point during PPI therapy; patients may present with varying signs and symptoms from

symptomatic hypersensitivity reactions to non-specific symptoms of decreased renal function (e.g., malaise, nausea, anorexia); in reported case series, some patients were diagnosed on biopsy and in absence of extra-renal manifestations (eg, fever, rash or arthralgia); discontinue therapy and evaluate patients with suspected acute TIN

Hypomagnesemia and mineral metabolism

- Hypomagnesemia may occur with prolonged use (ie, >1 year); adverse effects may result and include tetany, arrhythmias, and seizures; in 25% of cases reviewed, magnesium supplementation alone did not improve low serum magnesium levels and the PPI had to be discontinued
- Hypomagnesemia may lead to hypocalcemia and/or hypokalemia and may exacerbate underlying hypocalcemia in at-risk patients; in most patients, treatment of hypomagnesemia required magnesium replacement and discontinuation of the PPI
- Consider monitoring magnesium and calcium levels prior to initiation of therapy and periodically while on treatment in patients with a preexisting risk of hypocalcemia (eg, hypoparathyroidism); supplement with magnesium and/or calcium as necessary. If hypocalcemia is refractory to treatment, consider discontinuing the PPI

Over-the-counter use

- Avoid otc use and see healthcare professional if experiencing trouble or pain swallowing food, vomiting with blood, or bloody or black stools, heartburn with lightheadedness, sweating or dizziness, chest pain or shoulder pain with shortness of breath; sweating; pain spreading to arms, neck or shoulders; or lightheadedness, frequent chest pain; these may be signs of a serious condition; see healthcare professional

Pregnancy & Lactation

Pregnancy

Available data from published observational studies overall do not indicate an association of adverse pregnancy outcomes with lansoprazole treatment; estimated background risk of major birth defects and miscarriage for the indicated populations are unknown

Available data from published observational studies failed to demonstrate an association of adverse pregnancy-related outcomes and lansoprazole use; methodological limitations of these observational studies cannot definitely establish or exclude any drug-associated risk during pregnancy

Not recommended in pediatric patients <2 years; nonclinical studies in juvenile rats with lansoprazole have demonstrated adverse effect of heart valve thickening

No adverse effects on embryo-fetal development occurred in studies performed in pregnant rats at oral lansoprazole doses up to 150 mg/kg/day (40 times the recommended human dose [30 mg/day] based on body surface area) administered during organogenesis and pregnant rabbits at oral lansoprazole doses up to 0 mg/kg/day

(16 times the recommended human dose based on body surface area) administered during organogenesis

Lactation

There is no information regarding presence of lansoprazole in human milk, effects on breastfed infant, or on milk production; however, lansoprazole and its metabolites are present in rat milk; developmental and health benefits of breastfeeding should be considered along with mother's clinical need for therapy and any potential adverse effects on breastfed child from treatment or from underlying maternal condition

Pregnancy Categories

A: Generally acceptable. Controlled studies in pregnant women show no evidence of fetal risk.

B: May be acceptable. Either animal studies show no risk but human studies not available or animal studies showed minor risks and human studies done and showed no risk.

C: Use with caution if benefits outweigh risks. Animal studies show risk and human studies not available or neither animal nor human studies done.

D: Use in LIFE-THREATENING emergencies when no safer drug available. Positive evidence of human fetal risk.

X: Do not use in pregnancy. Risks involved outweigh potential benefits. Safer alternatives exist.

NA: Information not available.

Pharmacology

Mechanism of Action

Proton pump inhibitor; binds to H⁺/K⁺-exchanging ATPase (proton pump) in gastric parietal cells, resulting in suppression of basal and stimulated acid secretion

Absorption

Bioavailability: 81-91%; decreased 50-70% if given 30 min after meals

Peak plasma time: 1.7 hr; food increases time to 3.7 hr

Duration (at steady state): >24 hr (PUD, esophagitis); 40 hr (Zollinger-Ellison syndrome)

AUC: Food decreases AUC by 50%

Onset

- Gastric acid suppression: 1-3 hr
- PUD: 1 week (initial); 4-8 days (peak)
- Esophagitis: 1-4 weeks (initial); 8 weeks (peak)

Distribution

Protein bound: 97-99%

Vd: 14-18 L

Metabolism

Metabolized by hepatic CYP2C19; slow metabolizers are deficient in CYP2C19 enzyme and can have plasma concentration increase of 5-fold or higher

Gastric parietal cells: Acidic pH converts lansoprazole to its active sulfenamide metabolites

Active metabolites: Cyclic sulfenamide and disulfide metabolite

Inactive metabolites: 5-hydroxy-lansoprazole, sulfide metabolite, omeprazole sulfone, sulfone metabolite, hydroxysulfide metabolite, hydroxysulfone metabolite

Enzymes inhibited: CYP2C19

Elimination

Half-life: 0.9-1.5 hr

Dialyzable: No

Total renal clearance: 517 mL/min

Total body clearance: 0.7 L/hr/kg

Excretion: feces (bile), 67%; urine, 33%

Administration

Oral Administration

Administer before meals

Contains enteric coated granules (acid labile); do not chew or crush; swallow capsule/tablet whole; do not chew, crush, or split

Do not chew orally disintegrating tablets

IV Preparation

Inject 5 mL sterile water for injection (SWI) into vial and gently dissolve to obtain a 6 mg/mL solution; can be stored at 77°F (25°C) for 1 hr

Dilute in 50 mL NS, LR, or D5W; resulting solution can be stored at 77°F (25°C) for 24 hr (NS, LR) or 12 hr (D5W); refrigeration not necessary.

Alternatively, can be reconstituted directly into 50 mL NS, LR, or D5W using Baxter's MINI-BAG Plus Container

Prime in-line filter as directed in manufacturer's package insert

IV Administration

In-line filter must be used (supplied)

Administer over 30 min

Flush IV line with NS, LR, or D5W before and after use

Do not administer with other drugs

Formulary

Adding plans allows you to compare formulary status to other drugs in the same class.

Formulary information is available for health plans only in the United States.

Medscape prescription drug monographs are based on FDA-approved labeling information, unless otherwise noted, combined with additional data derived from primary medical literature.
